



Statistical Analysis of the Evolution of Professional Tennis

Raicu Maria-Camelia
Rosca Eduard-Mitrut
Regus Albert



Introduction

Professional tennis has undergone significant changes over the past decades due to:

- improvements in training and sports science,
- modifications of court surfaces and playing conditions,
- evolving player profiles and strategies.



Tennis Terminology

Serve: The fundamental overhead shot used by a player to start every single point in a match.

Ace: A perfectly placed legal serve that the opponent is completely unable to touch with their racket, that leads to a winning point for the serving player.

Fault and Double Fault: A fault is a serve that fails to land in the correct service box; two in a row is a double fault, only then leading to a lost point for the serving player.

Surfaces: Refers to the different types of court materials on which tennis is played, like clay, grass, or hard court, each having its own advantages and drawbacks.

ATP: Association of Tennis Professionals (Men's pro tennis)

WTA: Women's Tennis Association (Women's pro tennis)



Dataset

ATP/WTa Match Dataset

- Source: *ATP/WTa Tennis Data Repository*
- Coverage:
 - Professional ATP and WTA matches
 - Multiple decades of historical data
- Size:
 - ~178.000 ATP matches
 - ~195.500 WTA matches (Although many incomplete)



Dataset

- Data type:
 - **Match-level observations:**
 - Aces, double faults, first/second serve points won
 - Match duration (minutes), score, number of sets, surface
 - **Player attributes:**
 - Age, height, handedness, ranking
 - **Contextual information:**
 - Tournament level, round, surface, league (ATP/WTa)



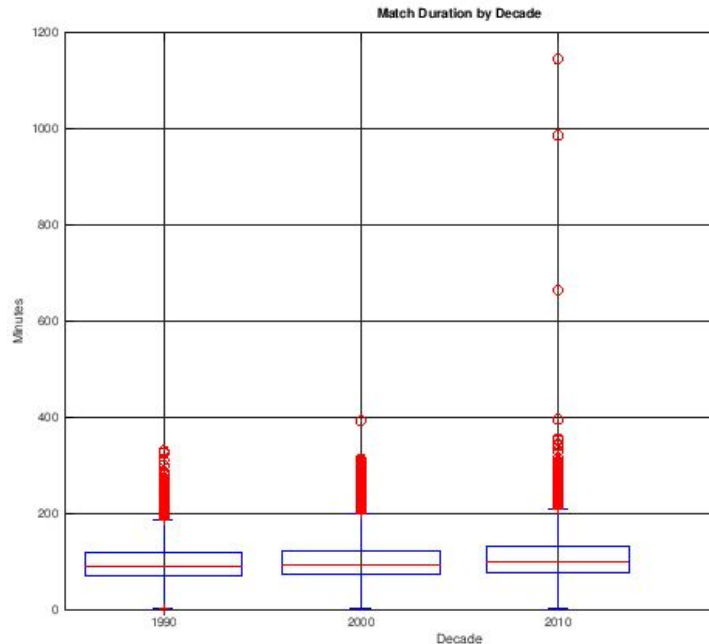
Objectives

Compare tennis characteristics across different eras

Identify statistically significant changes in:

- match performance
- player characteristics
- gender ATP vs. WTA

Match Performance

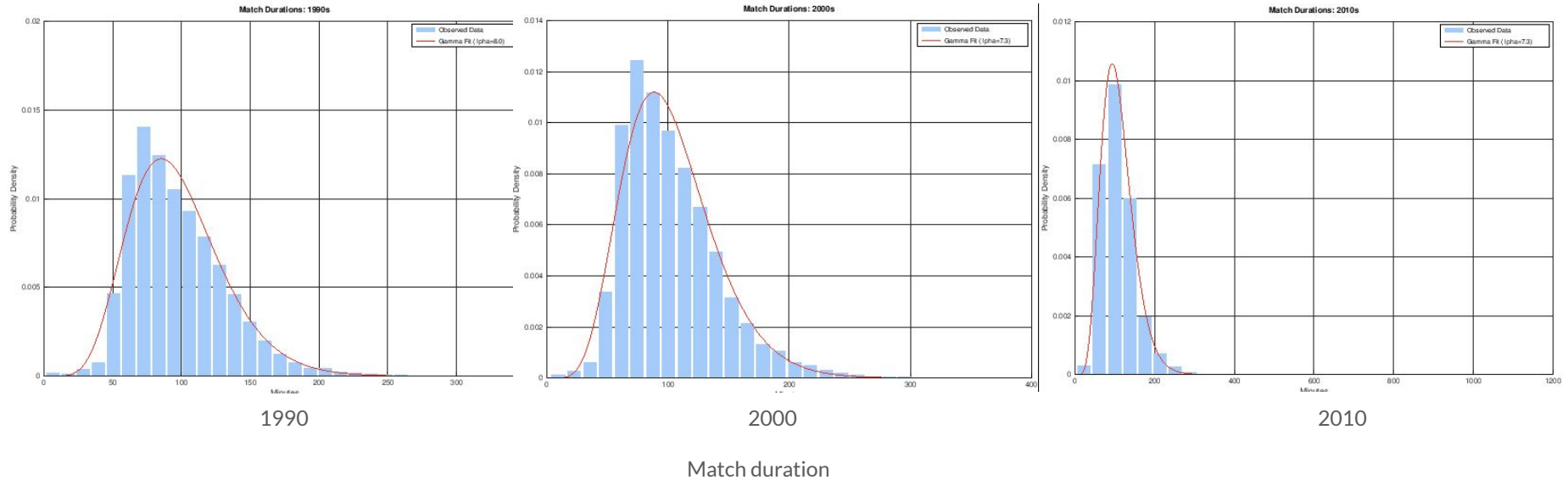


On ATP match performance

Descriptive Statistics: Match Duration

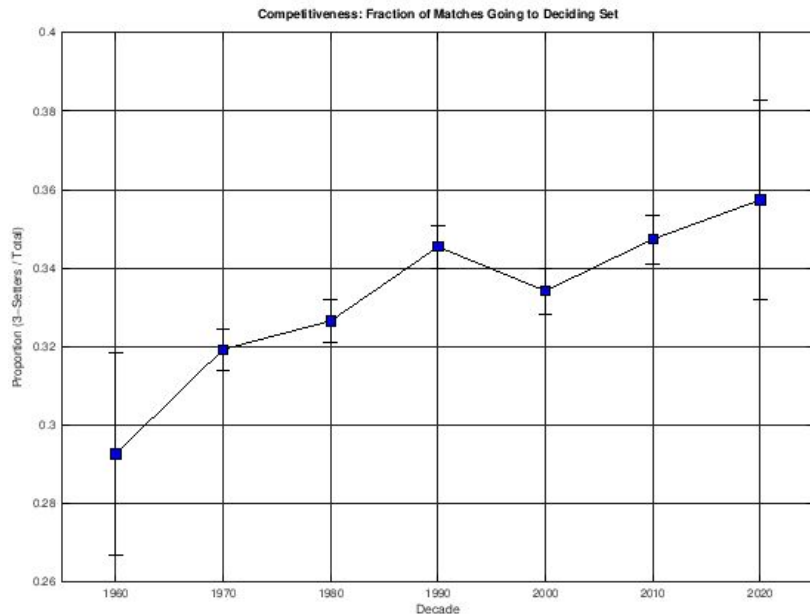
- Comparison of match duration distributions by decade
- Medians, variability, and extreme values highlighted
- Outliers represent exceptionally long matches

Match Performance



- while the *scale and variability* of match duration have changed over time, the *underlying distributional structure* remains similar.

Match Performance

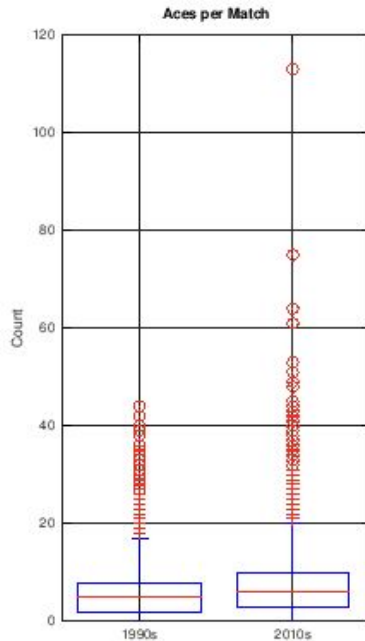


Inferential Statistics: Match Competitiveness

- Sample proportion of **three-set matches** and construct confidence intervals for the true proportion

Decade	Proportion	LB	UB	Matches (n)
1970s	0.3193	0.3139	0.3246	29566
1980s	0.3265	0.3211	0.3319	29066
1990s	0.3454	0.3400	0.3508	29593
2000s	0.3342	0.3283	0.3400	24820
2010s	0.3474	0.3411	0.3536	22481

Match Performance



Hypothesis Testing: Evolution of Serving - Aces per match

T-Test: Aces per Match

Hypotheses:

- H0: Mean number of aces per match is the same in both decades
- H1: Mean number of aces per match differs between decades

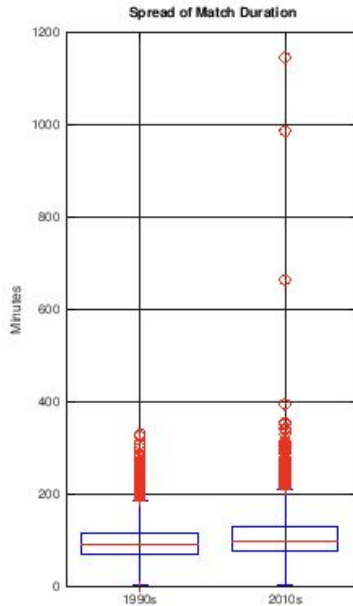
Results:

- Mean (1990s): **5.68**
- Mean (2010s): **7.24**
- T-statistic: **-34.63**
- P-value: **< 0.001**

Conclusion:

Reject H0: The increase in aces is statistically significant.

Match Performance



Hypothesis Testing: Variability in Match Duration

F-Test for Equality of Variances

Question:

- Is match performance more variable today than in the past?

Metric:

- Variance of ace counts per match

Hypotheses:

- H0: Variances are equal
- H1: Variances differ

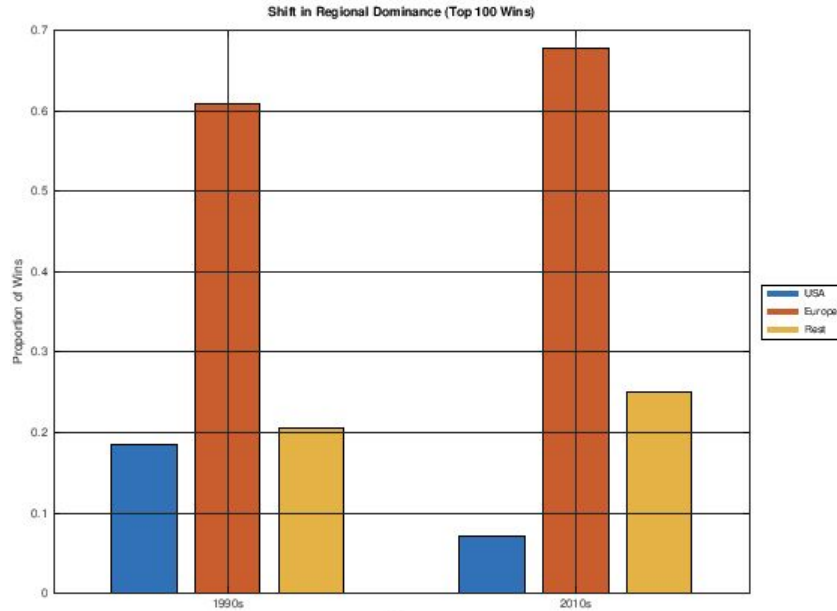
Results

- Variance (1990s): **22.77**
- Variance (2010s): **33.80**
- Variance ratio (2010s / 1990s): **1.48**
- P-value: **< 0.001**

Conclusion:

Reject H0. Variability has significantly increased.

Match Performance



Chi-Square Test: Geographic Shift in Tennis - **Proportion of wins**

Focus:

- Top 100 ranked players only

Regions:

- USA
- Europe
- Rest of the World

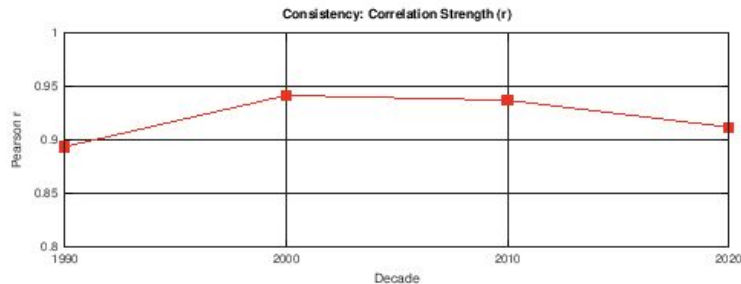
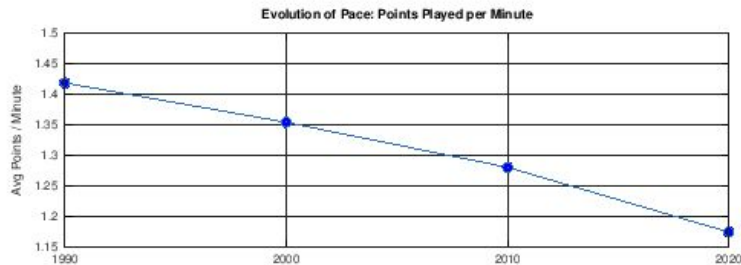
Hypotheses:

- H0. The geography of tennis has not changed significantly.
- H1. The geography of tennis has changed significantly.

Result

- Chi-square statistic: 1455.84
- P-value: < 0.001

Match Performance



Correlation Analysis: Pace of Play

Variables:

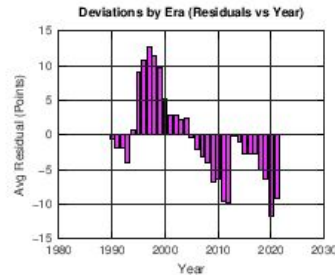
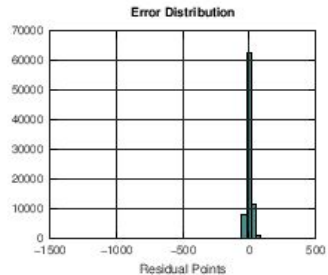
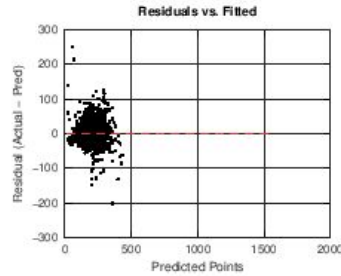
- Match duration (minutes)
- Total serve points played ($w_svpt + l_svpt$)

Method:

- Pearson correlation (by decade)
- Simple linear regression:
 - Slope = **points played per minute** ("pace")

Decade	Correlation (r)	P-Value	Slope (Pts/Min)	Matches
1990s	0.8940	0.0e+00	1.42	28363
2000s	0.9420	0.0e+00	1.35	28539
2010s	0.9374	0.0e+00	1.28	25852
2020s	0.9123	0.0e+00	1.17	1818

Match Performance



OLS Regression: Total Points vs Match Duration

- least square regression
- residuals vs fitted - no strong systematic pattern, suggesting that the linearity assumption is reasonable.
- error distribution - approximately symmetric and centered around zero
- avg residuals vs year - small but systematic deviations across eras

Results:

- **R sq = 0.847**
- **Slope = 1.33 points/minute**

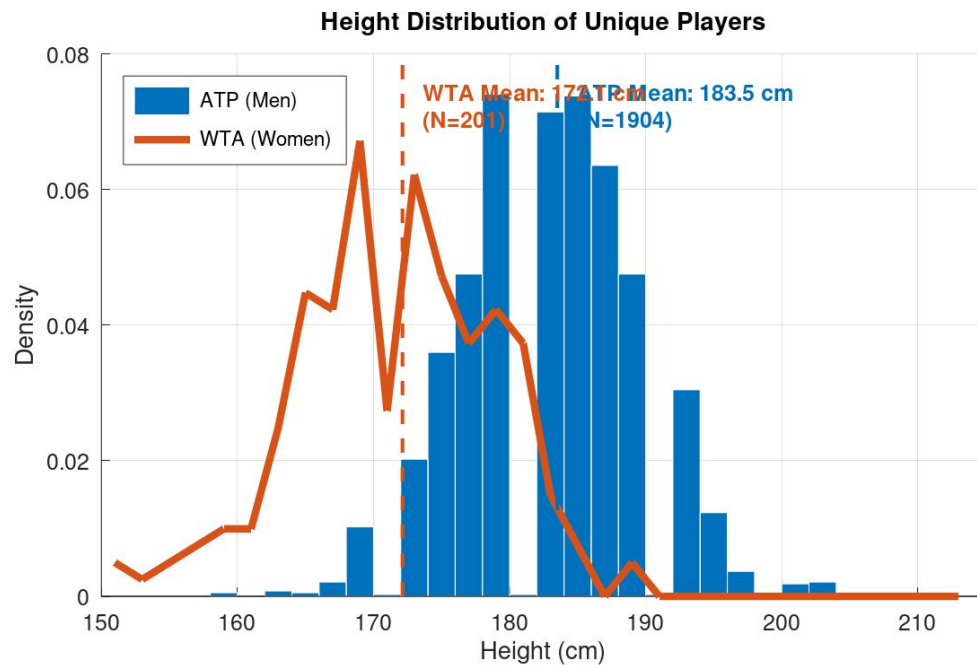


Player Characteristics

Growth trends: Investigating the evolution of player physiology to see growth patterns and differences over decades

Evolution of the handedness: Analyzing if the handedness changed over time and if it's statistically significant. Investigating if the handedness is affecting serving performance.

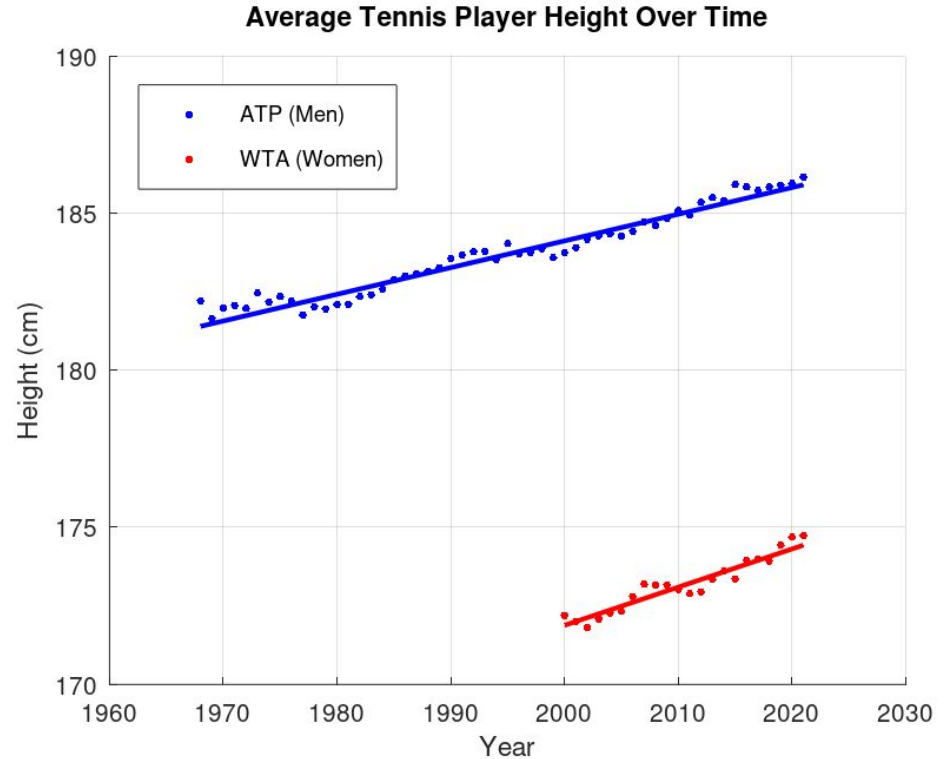
Height Distribution (ATP, WTA)



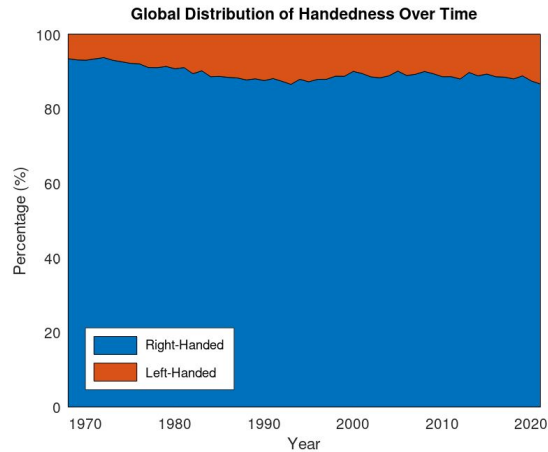
Growth Trends

The analysis of player height over time was conducted using linear regression and reveals a statistically significant upward trend for both male (ATP) and female (WTA) professional tennis players.

Men are growing taller by 0.85 cm / decade, while women are growing at a faster rate of 1.21 cm / decade resulting that players are getting taller over time.



Distribution of handedness over time



The analysis shows that the professional tennis players handedness shifted over time.

The statistical method used was the Chi-Square test to test if the handedness changed over time



Handedness over time

Chi-Square test:

$$\chi^2 = 73$$

$$p < 0.001$$

Trend: There has been an increase in the number of the left-handed players, rising from 7.7% in the 70' to 12.8% in the 2020'.

This indicates that left-handed players are becoming more common in the modern game.



Handedness on Aces

Does handedness affect serving performance? Is there a significant difference in the rate of Aces between Left-handed and Right-handed players?

The Test: **Welch's Two-Sample T-test**

League	ATP		WTA	
Hand	Left	Right	Left	Right
Mean	7.09%	7.21%	3.85%	3.72%



Handedness on Aces

Results ATP:

T-Statistic: -2.92

P-Value: 0.0035

Results WTA:

T-Statistic: 2.34

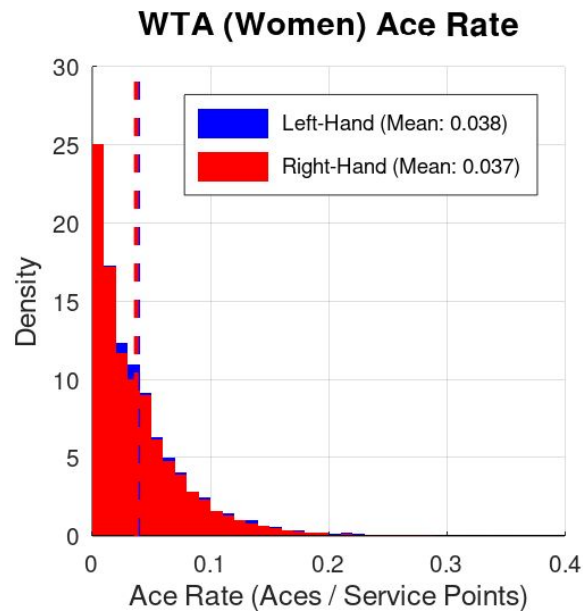
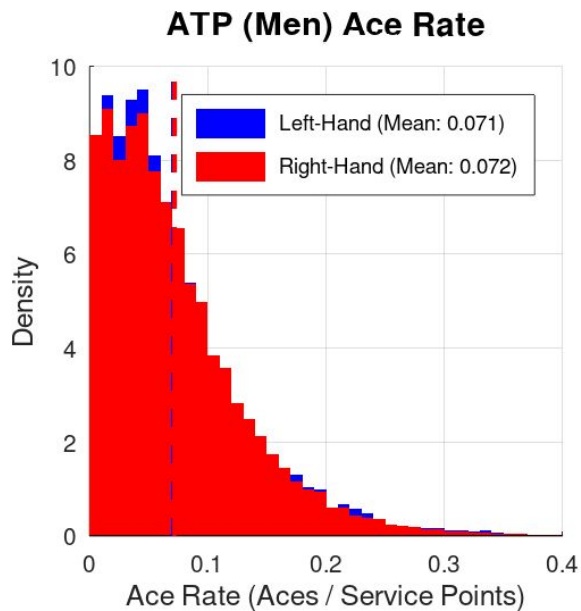
P-Value: 0.0192

Conclusion:

- in ATP: Right-handed tennis players have a statistically significant higher ace rate compared to left-handed players.
- in WTA: Left-handed players have a statistical higher ace rate compared to right handed players.

This demonstrates that there is no 'Left-hand advantage' in men's tennis, whereas a significant advantage does exist in women's tennis.

Handedness on Aces





ATP (Men) vs. WTA (Women) Comparative Analysis

The Ace Gap: Visualizing the difference in serve dominance and testing its statistical significance

Physical Drivers: Regression analysis on how height correlates with aces in both leagues.

Match Competitiveness Theory: Testing the hypothesis that lower serve dominance leads to more deciding sets.

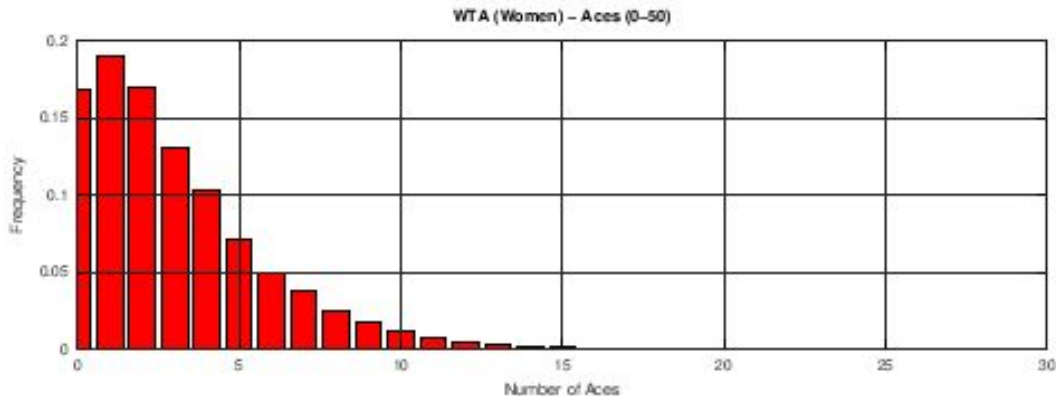
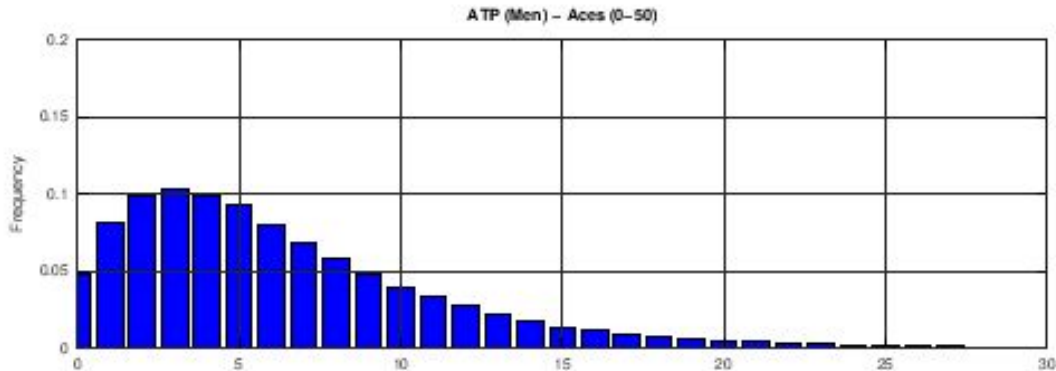
Case Study: Testing my roommates theory that Simona Halep likes to wear her opponents down

Ace Gap

Men (ATP): Distribution is "heavy-tailed." A significant population of players averages 10+ aces/match.

Women (WTA): Tightly clustered near zero. High ace counts are rarer.

What that might mean: The serve is a primary weapon in the men's game, whereas the women's game relies more on return and rallies.





Ace Gap

Could the gap in aces just be random chance? Is the difference real?

The Test: Welch's Two-Sample T-Test.

The Result:

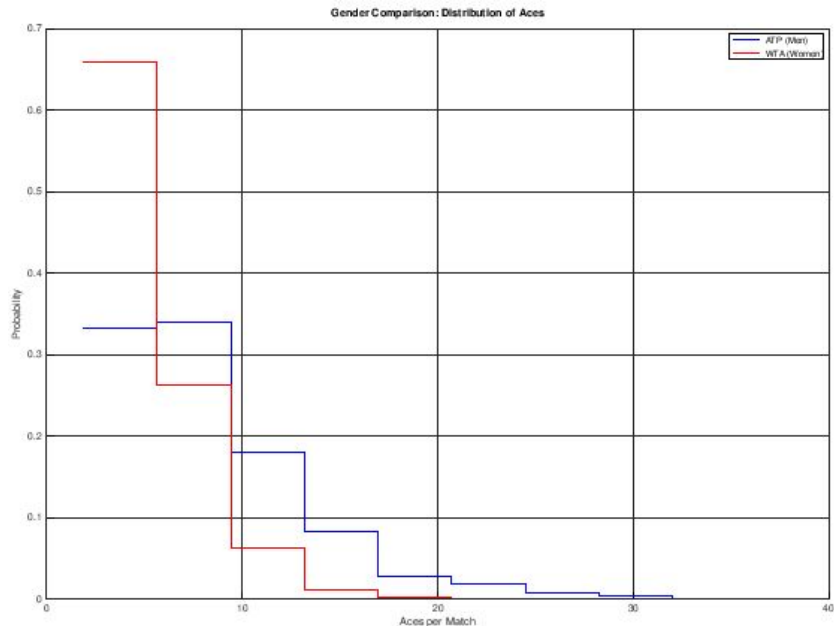
- **T-Statistic:** 115.25
- **P-Value:** < 0.00001
- **CI:** [3.4011, 3.5187] (3.4011, 3.5187)

Conclusion: The difference in serving performance is statistically significant and fundamental to the respective leagues, not a product of luck.

Ace Gap

Metric	ATP	WTA
Mean Aces	6.50	3.04
Difference	+114%	

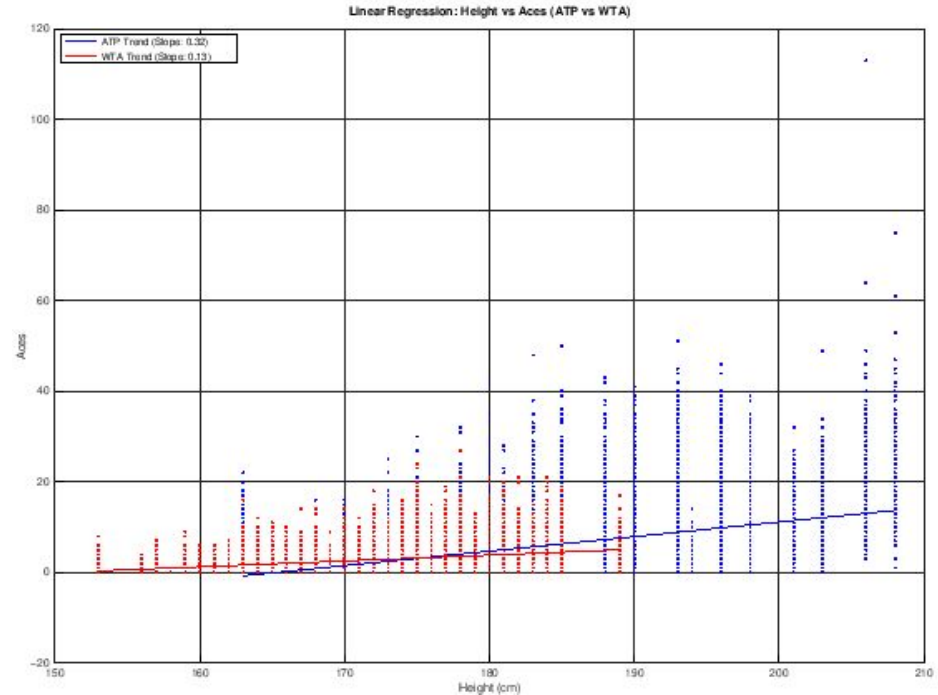
- Men hit 2.14x more aces per match
- Reflects serve speed advantage (~120 mph vs ~95 mph avg)



Height Analysis

- **ATP:** Slope ~ 0.32 (10cm would add ~ 3.2 aces/match).
- **WTA:** Slope ~ 0.13 (10cm would add ~ 1.3 aces/match).

Conclusion: Height has 2.5x stronger effect on men's serving. In the women's game, being tall offers diminishing returns compared to men.





Match Competitiveness Theory

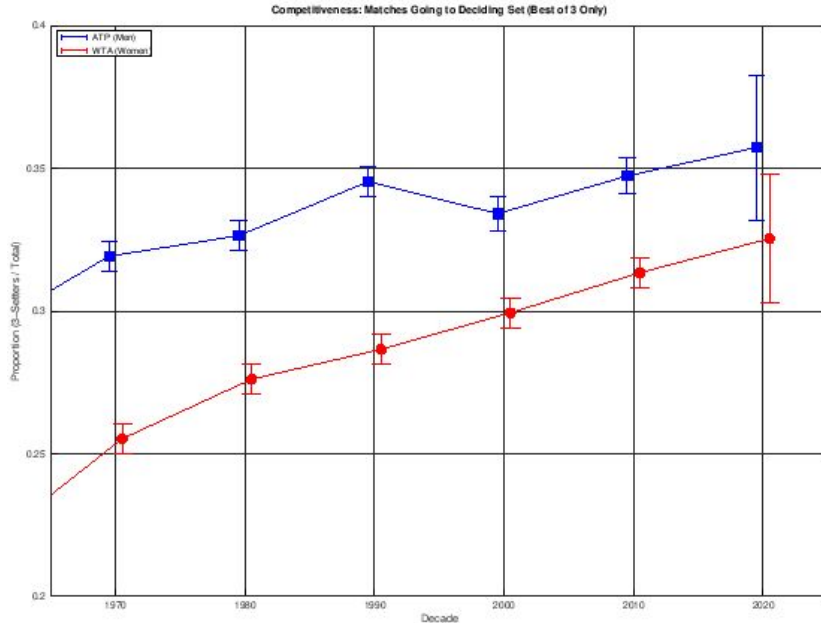
Hypothesis:

- Women's matches should go to a deciding set MORE OFTEN than men's matches because of weaker serves and less aces that create more competitive, back-and-forth contests.

Logical Reasoning:

- Lower ace rate (3.04 vs 6.50) -> fewer "free points" for the serving player
- More break opportunities (16.46 vs 13.93 BP) -> more service breaks
- Less serve dominance -> closer score lines -> more 3-set matches

Match Competitiveness Theory



What the data actually shows:

- The Hypothesis was **WRONG**, possibly because of Momentum swings, Mental factors and serve variability.

Decade	ATP	WTA	Difference
1980s	32.6%	27.6%	-5%
1990s	34.5%	28.7%	-5.8%
2000s	33.4%	29.9%	-3.5%
2010s	34.7%	31.4%	-3.3%
2020s	35.7%	32.5%	-3.2%



Case Study: Simona Halep

My roommates Hypothesis:

- Simona Halep is a defensive player that extends points and wears her opponents down until they are physically and mentally exhausted.

What the data shows:

- her matches are statistically shorter (89 min) than the WTA average (96 min) (T-Test: $p \sim 0.01$).
- Halep suppresses opponent aces more often than the WTA average (T-Test: $p < 0.001$).
 - WTA Average: ~ 2.5 Aces/Match
 - Halep's Opponents: ~ 1.8 Aces/Match

Which leads us to the conclusion that she returns serves so well that she breaks opponents quickly, making matches shorter at the same time.



Conclusion

- Serve dominance has increased significantly over time
- Despite stronger serving, matches have become physically slower and more variable
- Player physiology has evolved, with athletes becoming taller and more frequently left-handed.
- A significant and statistically proven Ace Gap exists between the ATP and WTA tours.
- Height provides a greater serving advantage for male players than for female players.
- The theory that lower serve dominance in the WTA leads to more competitive matches is incorrect.
- Data analysis reveals that serve returning, not just serving, is crucial for match outcomes.
- Statistical analysis can challenge and disprove long-held beliefs and anecdotal theories in tennis.